CS32(O)

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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SUPPLEMENTARY SEMESTER EXAMINATIONS – JULY 2023

Program	:	**B.E :- Computer Science and Engineering**	**Semester**	:	**III**
Course Name	:	**Data Structures**	**Max. Marks**	:	**100**
Course Code	:	**CS32(O)**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT – I

1. a) Write a C function to find the simple sparse transpose of a matrix. CO1 (06)

   b) Explain any two dynamic memory allocations functions in C with syntax and Write a function for dynamically allocating memory for a 2D array and explain. CO1 (06)

   c) Define a C structure to represent the terms of a polynomial and use an array of that structure to store the following polynomials: CO1 (08)

   A(x)= 9x^7 + 5x^3 – 2x^2 – 10

   B(x)= 14x^5 + 6x^3 + 7x + 3

   Illustrate the array representation of these two polynomials with the help of block diagram.

2. a) Illustrate how a two dimensional array is allocated dynamically, with a suitable example. CO1 (06)

   b) Write necessary C functions to add two polynomials represented using an array. CO1 (08)

   c) Write a recursive C function to find the length of a string. CO1 (06)

UNIT – II

3. a) Write C function which demonstrate queue operations. CO2 (06)

   b) Write a C function to reverse a string using stack. CO2 (08)

   c) Convert the infix expression a/b – c + d * e – a * c into postfix expression. Write a function to evaluate the postfix expression and trace that for given data a=6, b=3, c=1, d=2, e=4. CO2 (06)

4. a) Write a C function that evaluates a **prefix** expression. (Hint: Use queue) CO2 (10)

   b) Write a C function for stackFull() for multiple stacks. If the i^{th} stack is full and the $(i+1)^{th}$ stack has free memory space, then increase the i^{th} stack memory space to store a new element, by acquiring the memory space from $(i+1)^{th}$ stack. CO2 (10)

UNIT – III

5. a) Write a C function to display a doubly linked list in backward direction. CO3 (08)

   b) List the advantages of linked lists over arrays. CO3 (06)

   c) What is linked list? Write a C routine to insert an integer. CO3 (06)

6. a) Write C functions for push() and pop() operations for a linked stack. CO3 (06)

   b) Represent the following two polynomials using singly linked lists. CO3 (06)

   A(x)= 5x^6 + 6x^4 –4x + 10

   B(x)= 4x^7 + 2x^3 + 7x^2 – 9

   c) Write a C function to delete a node from a doubly linked circular list with a header node. Also illustrate with an example. CO3 (08)

UNIT – IV

7.	a)	Define with an example each of the following: i) Binary tree ii) Full binary tree iii) Complete binary tree.	CO3	(06)
	b)	Describe Satisfiability problem. Justify how it can be solved using binary tree with a suitable example.	CO3	(08)
	c)	Specify the min heap and max heap with detail descriptions.	CO4	(06)
8.	a)	What is satisfiability algorithm? Explain it.	CO4	(06)
	b)	Explain the winner and loser trees with a suitable examples.	CO4	(08)
	c)	Construct a binary tree for the given inorder and preorder traversals. Inorder traversal : b d h f a e g c Preorder traversal: a b d f h c e g	CO4	(06)

UNIT – V

9.	a)	Build an AVL tree with the following values: 15, 20, 24, 10, 13, 7, 30, 36, 25 Show step by step tree construction highlighting the rotations performed.	CO5	(10)
	b)	Write functions for i) DFS traversal ii) BFS traversal.	CO5	(10)
10.	a)	Describe the height biased leftist trees with suitable examples.	CO5	(10)
	b)	Write necessary C functions for breadth first search of a graph represented as an adjacency list.	CO5	(10)

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